

1. Data Construction

Antioxidant Food Table

Carlsen et al. 2010 (CC-BY)
138-page PDF supplement

Structured Extraction

word bounding boxes
+ statistical validation

Modelling Table

3,135 records · $\log(1+\text{FRAP})$
product text + category

2. Leakage-Aware Split Design

Name Normalisation

39.4% of records share
a product name

Split
× 5 seeds

Grouped by Name

fit / val / test
(leakage-free)

Random Split

fit / val / test
(37% duplicates in test)

3. Model Development

Baselines

global median
category median
memorization
(duplicate lookup)

Classical ML

TF-IDF + category
Ridge · k-NN · RF
XGBoost · LightGBM · MLP

Deep Learning

char-CNN · BiLSTM
+ category embedding
→ 192-d representations

ML-DL Integration

embeddings →
XGBoost / Ridge
8-model fusion

4. Evaluation and Diagnostics

Leakage-Aware Scoring

R^2 · RMSE · Spearman ρ on $\log(1+\text{FRAP})$
5 paired seeds · y-permutation controls

Diagnostics

$\Delta R^2 = \text{random} - \text{grouped} \cdot \text{memorization}$
decomposition · text / category ablations